13. Wireshark

i) Packet Capture Using Wireshark

ii) Starting Wireshark

iii) Viewing Captured Traffic

iv) Analysis and Statistics & Filters

Step 1: Update System Packages

Open the terminal and update the package list to ensure the latest versions are available.

```
sudo apt update
```

Step 2: Install Wireshark

Install Wireshark using apt package manager.

```
sudo apt install wireshark
```

During installation, you might be prompted if non-superusers should be allowed to capture packets. Choose Yes to allow users in the Wireshark group to capture packets without root privileges.

Step 3: Add User to Wireshark Group

To allow capturing packets without sudo, add your user to the Wireshark group.

```
sudo usermod -aG wireshark $USER
```

- Log out and log back in for group membership to apply.
- Verify membership by running:

```
groups
```

and checking for wireshark in the output.

Step 4: Start Wireshark

You can launch Wireshark from the terminal or GUI.

Terminal command:

```
wireshark
```

Or find Wireshark in the Ubuntu application menu.

Step 5: Select Network Interface and Start Capture

Upon opening, you will see a list of available network interfaces.

Select the interface through which you expect traffic (e.g., eth0, wlan0).

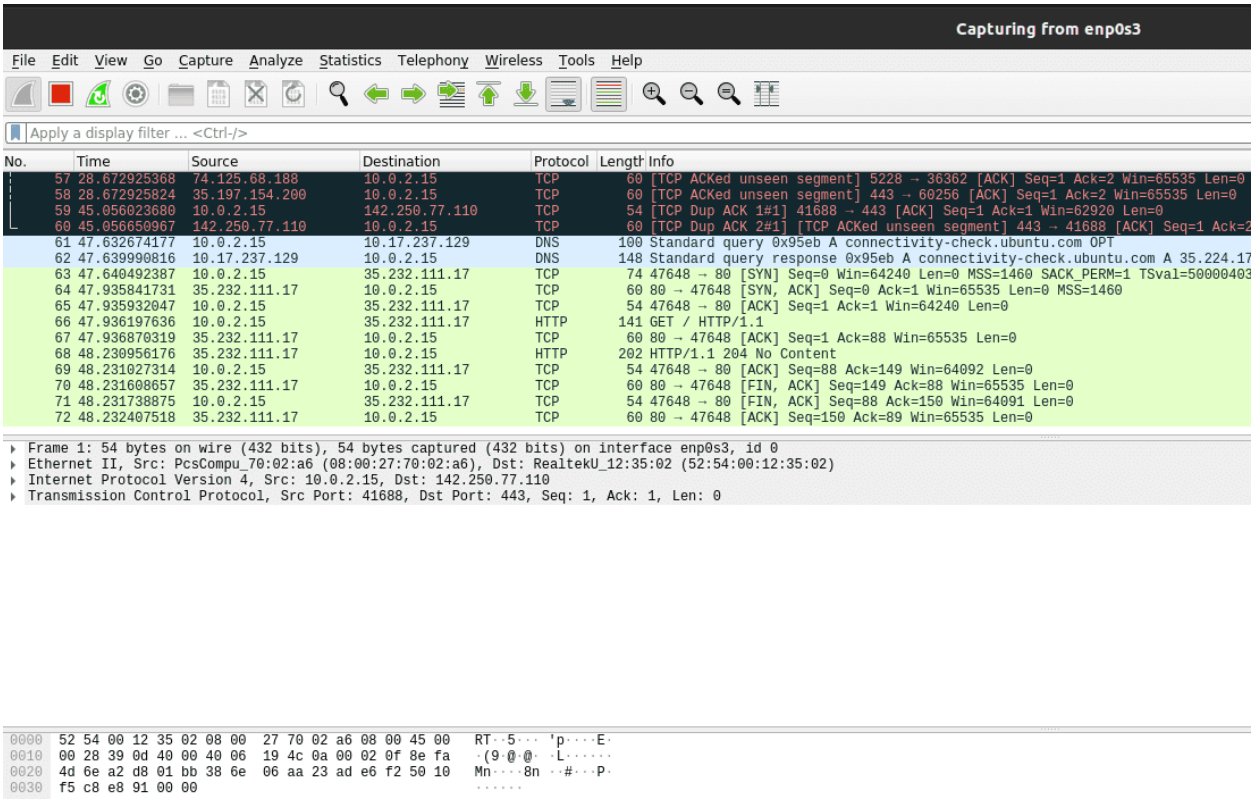
Click the blue shark fin button at the top to start packet capture.

Step 6: Viewing Captured Traffic

Packets appear live in real-time.

Columns show important details like No., Time, Source, Destination, Protocol, Length, and Info.

Click any packet to expand its detailed view with protocol layers like Ethernet, IP, TCP/UDP.

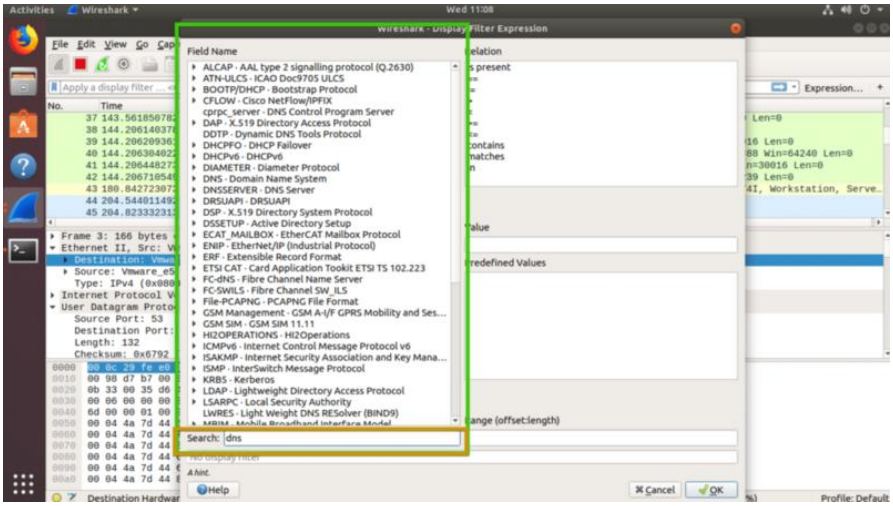


Step 7: Using Filters

Use the display filter box at the top to filter traffic by protocol or condition.

Examples:

- tcp to show only TCP packets.
- ip.addr == 192.168.1.5 to show traffic involving specific IP.
- http to filter HTTP traffic.
- Press Enter to apply the filter.

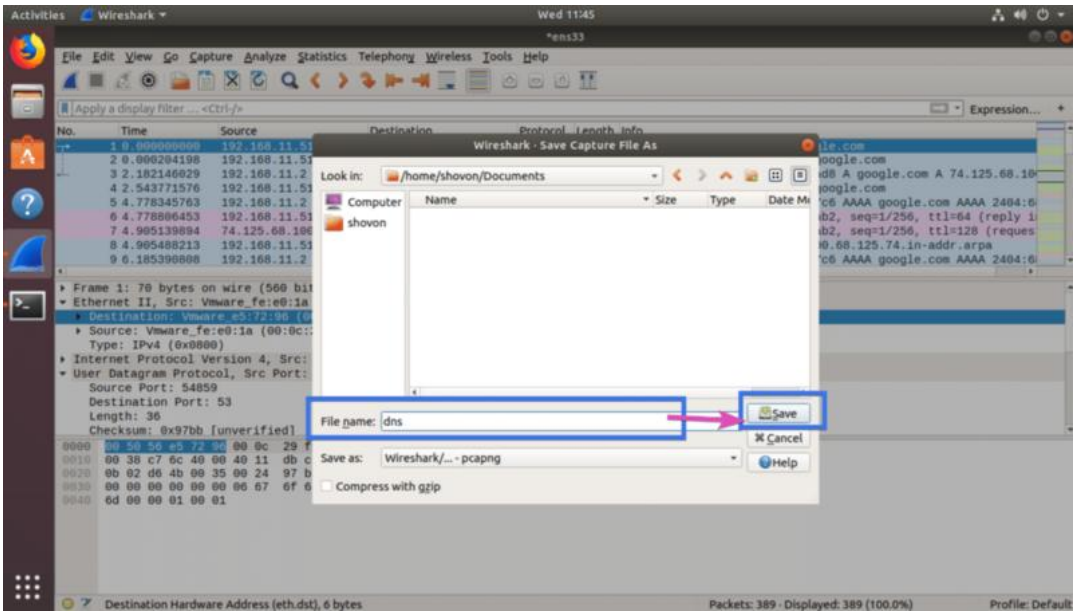


Step 8: Analyze Using Statistics Menu

- Navigate to the Statistics menu for summaries:
  - Protocol Hierarchy displays protocol distribution.
  - Conversations shows endpoints and traffic volume.
  - Endpoints lists all hosts seen in capture.
- Useful to understand traffic patterns and diagnose network issues.

Step 9: Save Captured Packets

- To save a capture, go to File > Save As...
- Choose location and filename with .pcapng extension.



Step 10: Stop Capture

- Click the red square stop button to end capturing.